

Research on Driving Factors of COVID-19 Protective Behaviours in Australia Based on Deep Learning

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Introduction

- Personal protective behaviours, especially face masks, played a crucial role in controlling the spread of COVID-19.
- Public health policies, such as mask mandates, significantly alter public compliance and the underlying behaviour drivers (Ryan et al. 2025, p3–5). Understanding their drivers is important for public health policy.

Background

Multiple Factors Shape Behaviour

Protective behaviours are influenced by demographic, psychological, and policy factors, and may also change with vaccination and COVID-19 case numbers.

Drives of mask mandates and protective behaviour changes in Australia

Protective behaviours are not fixed, previous study showed the role of face mask mandates in influencing the drivers of protective behaviours in Australia. (Ryan et al. 2025, 8–13)

MLP has shown potential in structured data

MLP(Multi-Layer Perceptron)s can achieve competitive or even stronger performance on structured data when appropriate regularization and parameter tuning are applied.

Research Questions

- How do vaccination rates and COVID-19 case numbers influence the protective behaviours in Australia?
- How does the performance of deep learning models compare to traditional models in predicting protective behaviours?

Imperial College London YouGov Behaviour Tracker dataset

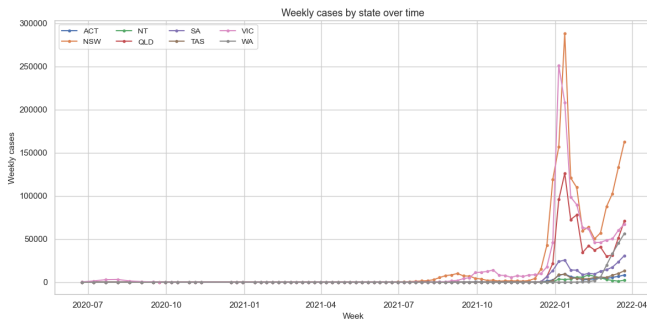
- Demographic and psychological factors (such as gender and age)
- protective behaviours (such as mask wearing and social distancing)
- Comorbidities
- other features.

Oxford COVID-19 Government Response Tracker dataset

- The state level response policies of governments around the world aimed at preventing the spread of COVID-19 from March 2020 to January 2023.

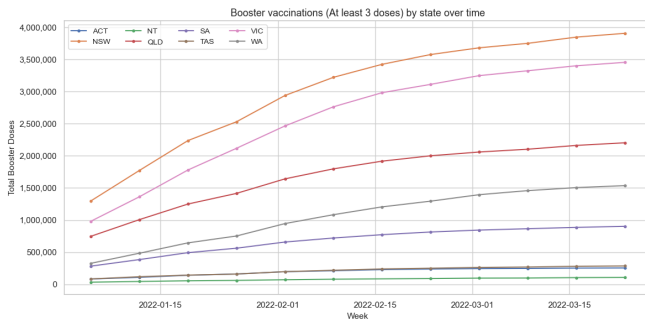
Data-Pandemic Case number

Consolidated dataset for COVID-19 case numbers, which compiles and verifies COVID-19 case data from Australian federal, state, and territory health departments. (covid19data.com.au 2023a)



Data-Vaccination Data

Vaccine dataset compiled by a third-party team to match the timing and states of the YouGov survey.
(covid19data.com.au 2023b)



Preprocessing

Government Mandates

This study calculated the 14-day rolling average of each state's index.

When the rolling average of a certain state reached 3 for the first time, it was considered that the state had actually entered a stable period of mandate intervention.

Removal of Features with High Missingness

Columns with missing values greater than 10,781 were considered not in the survey for a long period and were dropped from the dataset. (Ryan et al. 2025, 3)

Repairing Data Gaps Caused by Survey Changes

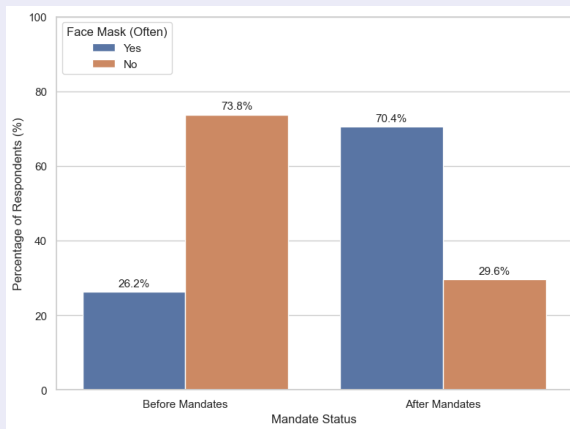
During the data exploration process, it was found that YouGov modified the survey between February 10 and October 18, 2021. Having verified that the imputation strategy does not introduce significant bias, this study located this time window using a date mask and filled in these missing values as independent 'N/A' categories.

Predictive Variables

Calculate the two-week average for the following three features and convert them into binary variables:

Mask wearing, Protective behaviours and protective behaviours other than masks.

Class Imbalance and Resampling



To overcome class imbalance, this study introduced the random oversampler technique through `imblearn.pipeline`.

Methods-Cross Validation, metrics and Interpretation

Cross validation

- A single validation set is highly susceptible to the randomness of the initial data split.
- To prevent overfitting during model training, this study implemented a 5-fold cross validation.

Evaluation metrics

- AUC-ROC serves as the primary metric for model comparison and parameter tuning.
- Precision, Recall, and F1 score for a specific comparison of model performance differences.

Interpretation

- SHAP summary plots are applied to the best-performing model to evaluate both the magnitude and directional impact of each predictor.

Logistic Regression

- Regularization strength
- Regularization type

Classification Trees

- Maximum depth of the tree
- Minimum samples per leaf
- Minimum samples to split a node

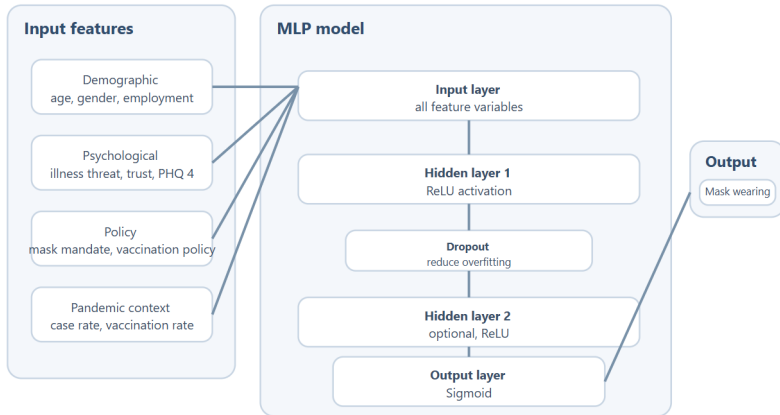
XGBoost

- Learning_rate
- Number of trees
- Maximum depth of each tree
- Number of features per split

Multilayer Perceptron

MODEL STRUCTURE

Proposed MLP architecture for predicting mask wearing



Training: 5 fold cross validation on the training set

Evaluation: F1 score and ROC AUC on the testing set

Interpretation: SHAP values for the two best performing models

Current progress

Experiment on handling structural missing values

- Content: Compared retaining np.nan, imputing an “N/A” category, and dropping features.
- Findings: Dropping features altered predictions, while “N/A” imputation maintained model performance.
- Decision: Proceeding with “N/A” string imputation, as most evaluated models cannot handle np.nan directly.

Model tuning

- The model definition for the modeling phase is complete, and tuning is still in progress.

Model tuning (w5-6)

Fit four models: Logistic Regression, Classification Trees, XGBoost, and MLPs. For the tree based models, hyperparameters will be tuned with 5-fold cross validation on the training data, with AUC as the optimisation metric.

Model comparison (w6-7)

Compare the four models using AUC, Precision, Recall, Accuracy, and F1 score. The best models will be selected based on predictive performance.

Interpretation (w7-9/10)

Investigate the stable important features by SHAP values on the best performing models.